

MODULE-10 : SOFTWARE MANAGEMENT

GUIDED BY : VIPUL MISTRI

Q.1 Use the apt package manager to install the VLC media player on your Linux system, then verify the installation by running `vlc --version` in the terminal.

ANS :

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal
Aug 21 15:16
root@localhost: /usr/bin/python3.9 /usr/bin/dnf install -y vlc

intel-vpl-gpu-rt          x86_64          26.1.6-1.el9          epel          3.8 M
jxl-pixbuf-loader        x86_64          0.7.2-2.el9           epel          52 k
libproxy-bin             x86_64          0.4.15-35.el9         appstream    15 k
openh264                 x86_64          2.3.1-1.el9           epel-cisco-openh264 425 k
vlc-gui-skins2           x86_64          1:3.0.23-2.el9        epel          421 k
vlc-plugin-ffmpeg        x86_64          1:3.0.23-2.el9        epel          97 k
vlc-plugin-gnome         x86_64          1:3.0.23-2.el9        epel          17 k
vlc-plugin-notify        x86_64          1:3.0.23-2.el9        epel          17 k
vlc-plugin-visualization x86_64          1:3.0.23-2.el9        epel          32 k
vlc-plugins-extra        x86_64          1:3.0.23-2.el9        epel          689 k

Transaction Summary
-----
Install 114 Packages

Total download size: 83 M
Installed size: 284 M
Downloading Packages
(1/114): SDL2-2.26.0-1.el9.x86_64.rpm
(2/114): fftw-libs-double-3.3.8-14.el9.x86_64.rpm
(3/114): libmad-0.15.1b-33.el9.x86_64.rpm
(4/114): libproxy-bin-0.4.15-35.el9.x86_64.rpm
(5/114): librabbitmq-0.11.0-7.el9.x86_64.rpm
(6/114): google-noto-sans-mono-fonts-20201206-4.el9.noarch.rpm
(7/114): libusb-0.1.7-5.el9.x86_64.rpm
(8/114): libshaderc-2026.1-1.el9.x86_64.rpm
(9/114): libva-2.22.0-1.el9.x86_64.rpm
(10/114): libvdpau-1.5-1.el9.x86_64.rpm
(11/114): ocl-icd-2.2.13-4.el9.x86_64.rpm
(12/114): google-noto-serif-fonts-20201206-4.el9.noarch.rpm
(13/114): pcre2-utf16-10.40-6.el9.x86_64.rpm
(14/114): protobuf-lite-3.14.0-17.el9.x86_64.rpm
(15/114): qt5-qtbase-common-5.15.9-11.el9.noarch.rpm
(16/114): qt5-qtbase-5.15.9-11.el9.x86_64.rpm
(17/114): qt5-qtsvg-5.15.9-2.el9.x86_64.rpm
(18/114): qt5-qtxwayland-5.15.9-1.el9.x86_64.rpm
(19/114): qt5-qtdeclarative-5.15.9-3.el9.x86_64.rpm
(20/114): qt5-qtx11extras-5.15.9-1.el9.x86_64.rpm
(21/114): sdl12-compat-1.2.60-1.el9.x86_64.rpm
(22-23/114): qt5-qtbase-gui-5.15.9-11.el9.x86_64.rpm
32% [=====]

] 4.0 MB/s | 27 MB 00:14 ETA
30°C Mostly cloudy 3:16 PM 8/21/2026
```

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal
Aug 21 15:18
ismail@localhost:

libshaderc-2026.1-1.el9.x86_64
libssh2-1.11.1-9.el9.x86_64
libtiff-9.0.4-23.el9.x86_64
libuimg-1.14.12-1.el9.x86_64
libvdpau-1.5-1.el9.x86_64
lirc-libs-0.10.0-36.el9.x86_64
mbdttls-2.28.8-1.el9.x86_64
openh264-2.3.1-1.el9.x86_64
protobuf-lite-3.14.0-17.el9.x86_64
qt5-qtbase-gui-5.15.9-11.el9.x86_64
qt5-qtxwayland-5.15.9-1.el9.x86_64
schrodinger-1.0.11-26.el9.x86_64
speexdsp-1.2.0-5.el9.x86_64
svt-av1-libs-0.9.0-1.el9.x86_64
vlc-cli-1:3.0.23-2.el9.x86_64
vlc-libs-1:3.0.23-2.el9.x86_64
vlc-plugin-lua-1:3.0.23-2.el9.x86_64
vlc-plugin-pulseaudio-1:3.0.23-2.el9.x86_64
vlc-plugins-extra-1:3.0.23-2.el9.x86_64
xcb-util-image-0.4.0-19.el9.x86_64
xcb-util-wm-0.4.1-22.el9.x86_64
zeromq-4.3.4-2.el9.x86_64

libsodium-1.0.18-9.el9.x86_64
libswresample-free-5.1.9-1.el9.x86_64
libudfread-1.1.2-2.el9.x86_64
libusb-1.0.1.7-5.el9.x86_64
libvmaf-2.3.0-2.el9.x86_64
live555-2023.06.20-2.el9.x86_64
ocl-icd-2.2.13-4.el9.x86_64
openpgm-5.2.122-28.el9.x86_64
qt5-qtbase-5.15.9-11.el9.x86_64
qt5-qtdeclarative-5.15.9-3.el9.x86_64
qt5-qtx11extras-5.15.9-1.el9.x86_64
sdl12-compat-1.2.60-1.el9.x86_64
spirv-tools-libs-2026.1-1.el9.x86_64
vapoursynth-libs-57-4.el9.x86_64
vlc-gui-qt-1:3.0.23-2.el9.x86_64
vlc-plugin-ffmpeg-1:3.0.23-2.el9.x86_64
vlc-plugin-notify-1:3.0.23-2.el9.x86_64
vlc-plugin-visualization-1:3.0.23-2.el9.x86_64
vlc-plugins-video-out-1:3.0.23-2.el9.x86_64
xcb-util-keysyms-0.4.0-17.el9.x86_64
xerces-c-3.2.5-1.el9.x86_64
zimg-3.0.6-2.el9.x86_64

libspatialaudio-3.1-5.20200406gitd926a2e.el9.x86_64
libswscale-free-5.1.9-1.el9.x86_64
libunwind-1.8.0-4.el9.x86_64
libva-2.22.0-1.el9.x86_64
libvpl-1:2.17.0-1.el9.x86_64
lpcnetfreedv-0.2-11.el9.x86_64
opencore-amr-0.1.6-3.el9.x86_64
pcre2-utf16-10.40-6.el9.x86_64
qt5-qtbase-common-5.15.9-11.el9.noarch
qt5-qtsvg-5.15.9-2.el9.x86_64
rav1e-libs-0.3.1-2.el9.x86_64
soxr-0.1.3-11.el9.x86_64
srt-libs-1.4.4-1.el9.x86_64
vlc-1:3.0.23-2.el9.x86_64
vlc-gui-skins2-1:3.0.23-2.el9.x86_64
vlc-plugin-gnome-1:3.0.23-2.el9.x86_64
vlc-plugin-pipewire-3-1.el9.x86_64
vlc-plugins-base-1:3.0.23-2.el9.x86_64
vo-amrwbenc-0.13-18.el9.x86_64
xcb-util-renderutil-0.3.9-20.el9.x86_64
xvidcore-1.3.7-9.el9.x86_64
zvbi-0.2.35-15.el9.x86_64

Complete!
[root@localhost /]# vlc --version
VLC is not supposed to be run as root. Sorry.
If you need to use real-time priorities and/or privileged TCP ports
you can use vlc-wrapper (make sure it is Set-UID root and
cannot be run by non-trusted users first).
[root@localhost /]# su ismail
[ismail@localhost /]$ vlc --version
VLC media player 3.0.23 Vetinari (revision 3.0.23-2-0-g7912887dd)
VLC version 3.0.23 Vetinari (3.0.23-2-0-g7912887dd)
Compiled by mockbuild on buildhw-x86-18.rdu3.fedoraproject.org (Jan 19 2026 00:00:00)
Compiler: gcc version 11.5.0 20240719 (Red Hat 11.5.0-11) (gcc)
This program comes with NO WARRANTY, to the extent permitted by law.
You may redistribute it under the terms of the GNU General Public License;
see the file named COPYING for details.
Written by the VideoLAN team; see the AUTHORS file.
[ismail@localhost /]$
```

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Q.2 .Update all installed packages on your Linux system using the appropriate package manager command, and take a screenshot of the output showing at least one package being upgraded.

ANS :

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit
Activities Terminal
Aug 21 15:21
root@localhost: /usr/bin/python3.9 /usr/bin/dnf update

volume_key-libs x86_64 0.3.12-17.el9 appstream 150 k
vulkan-loader x86_64 1.4.341.0-1.el9 appstream 160 k
webkit2gtk3 x86_64 2.52.5-1.el9 appstream 29 M
webkit2gtk3-jsc x86_64 2.52.5-1.el9 appstream 8.9 M
wget x86_64 1.21.1-11.el9 appstream 783 k
which x86_64 2.21-30.el9 baseos 41 k
wireless-regdb noarch 2025.10.07-1.el9 baseos 16 k
wireplumber x86_64 0.5.12-1.el9 appstream 123 k
wireplumber-libs x86_64 0.5.12-1.el9 appstream 443 k
wpa_supplicant x86_64 1:2.11-2.el9 baseos 1.7 M
xdg-utils noarch 1.1.3-13.el9 appstream 72 k
xfsprogs x86_64 6.4.0-9.el9 baseos 1.1 M
xorg-x11-drv-libinput x86_64 1.0.1-4.el9 appstream 45 k
xorg-x11-drv-vmware x86_64 13.2.1-19.el9 appstream 62 k
xorg-x11-drv-wacom x86_64 1.0.0-5.el9 appstream 300 k
xorg-x11-drv-wacom-serial-support x86_64 1.0.0-5.el9 appstream 22 k
xorg-x11-server-Xorg x86_64 1.20.11-38.el9 appstream 1.5 M
xorg-x11-server-Xwayland x86_64 24.1.9-7.el9 appstream 1.0 M
xorg-x11-server-common x86_64 1.20.11-38.el9 appstream 34 k
yajl x86_64 2.1.0-25.el9 appstream 38 k
yelp x86_64 2:40.3-4.el9 appstream 866 k
yelp-libs x86_64 2:40.3-4.el9 appstream 113 k
yum noarch 4.14.0-37.el9 baseos 89 k

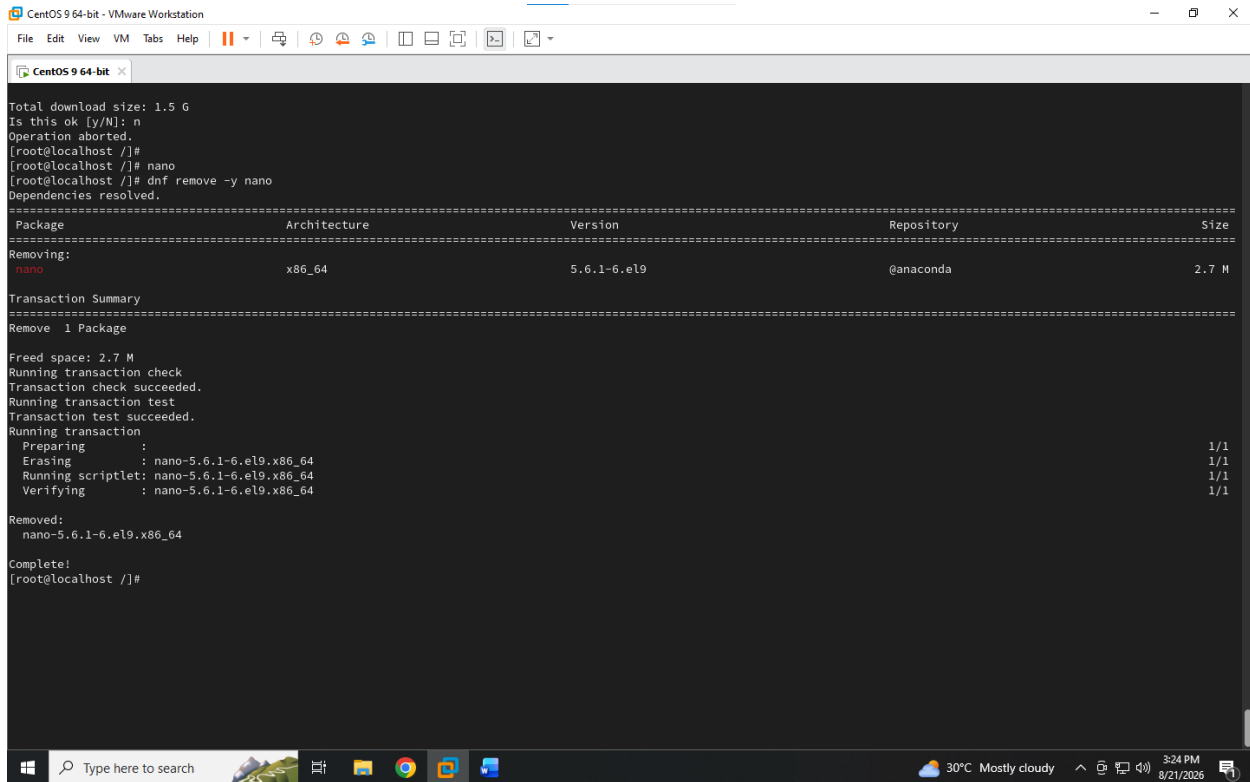
Installing dependencies:
adcli-selinux noarch 0.9.3.1-3.el9 baseos 19 k
cockpit-ws-selinux x86_64 365-1.el9 baseos 44 k
containers-common-extra x86_64 5:5.8-2.el9 appstream 5.2 k
fftw-libs-single x86_64 3.3.8-14.el9 appstream 935 k
grub2-tools-efi x86_64 1:2.06-134.el9 baseos 621 k
grub2-tools-extra x86_64 1:2.06-134.el9 baseos 939 k
gstreamer1-plugins-bad-free-libs x86_64 1.22.12-13.el9 appstream 772 k
kernel-core x86_64 5.14.0-737.el9 baseos 17 M
kernel-modules x86_64 5.14.0-737.el9 baseos 40 M
kernel-modules-core x86_64 5.14.0-737.el9 baseos 31 M
libbbltrace x86_64 1.5.8-10.el9 appstream 190 k
llvm-filestore x86_64 22.1.8-1.el9 appstream 13 k
openssl-fips-provider x86_64 1:3.5.7-3.el9 baseos 813 k
p11-kit-client x86_64 0.26.4-1.el9 baseos 268 k
python3-inotify noarch 0.9.6-25.el9 baseos 53 k
samba-core-libs x86_64 4.24.5-2.el9 baseos 491 k
```

MODULE-10 : SOFTWARE MANAGEMENT

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Q.3 Remove the nano text editor from your system using the package manager, then confirm it has been removed by attempting to run nano in the terminal. Hint: Use apt remove or yum remove depending on your Linux distribution.

ANS :



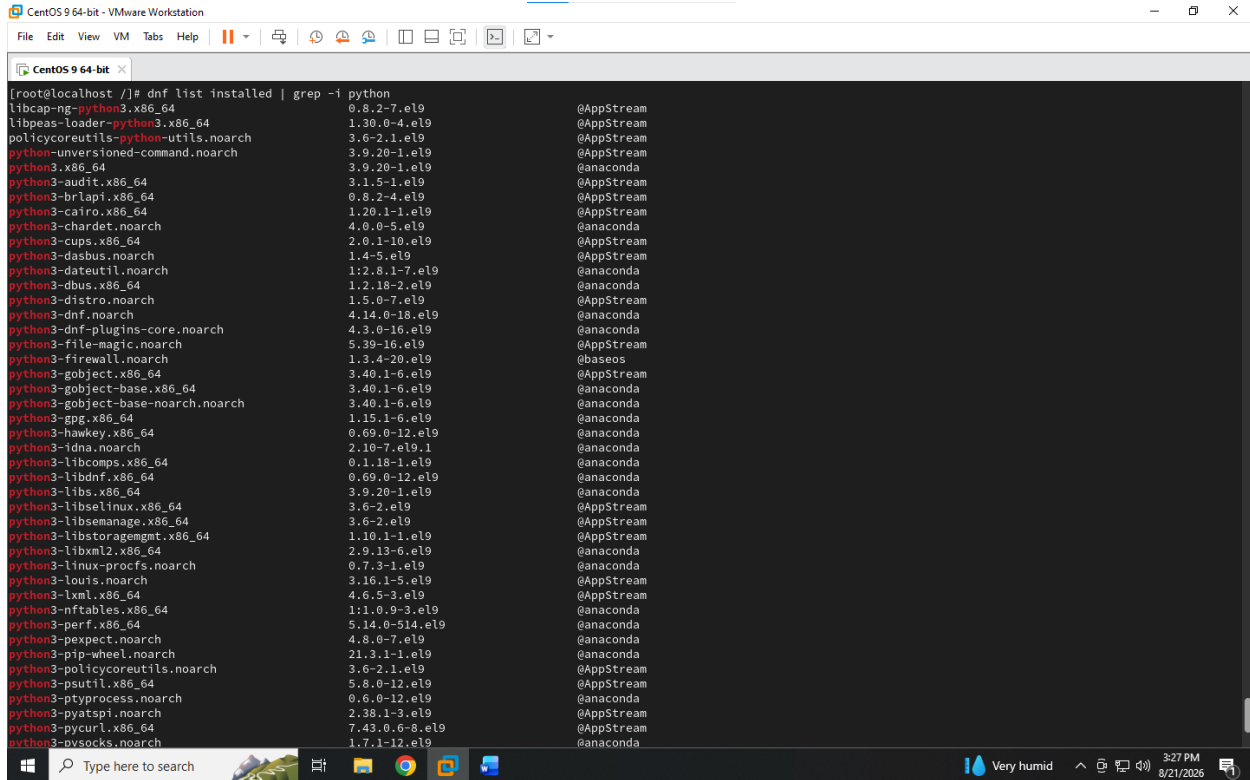
```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Total download size: 1.5 G
Is this ok [y/N]: n
Operation aborted.
[root@localhost ~]# nano
[root@localhost ~]# dnf remove -y nano
Dependencies resolved.
=====
Package Architecture Version Repository Size
=====
Removing:
nano x86_64 5.6.1-6.el9 @anaconda 2.7 M
Transaction Summary
=====
Remove 1 Package
Freed space: 2.7 M
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Erasing : nano-5.6.1-6.el9.x86_64 1/1
Running scriptlet: nano-5.6.1-6.el9.x86_64 1/1
Verifying : nano-5.6.1-6.el9.x86_64 1/1
Removed:
nano-5.6.1-6.el9.x86_64
Complete!
[root@localhost ~]#
```

MODULE-10 : SOFTWARE MANAGEMENT

GUIDED BY : VIPUL MISTRI

Q.4 List all installed packages on your system and filter the results to show only packages related to 'python'. Hint: Use grep to filter the output of your package manager's list command.

ANS :



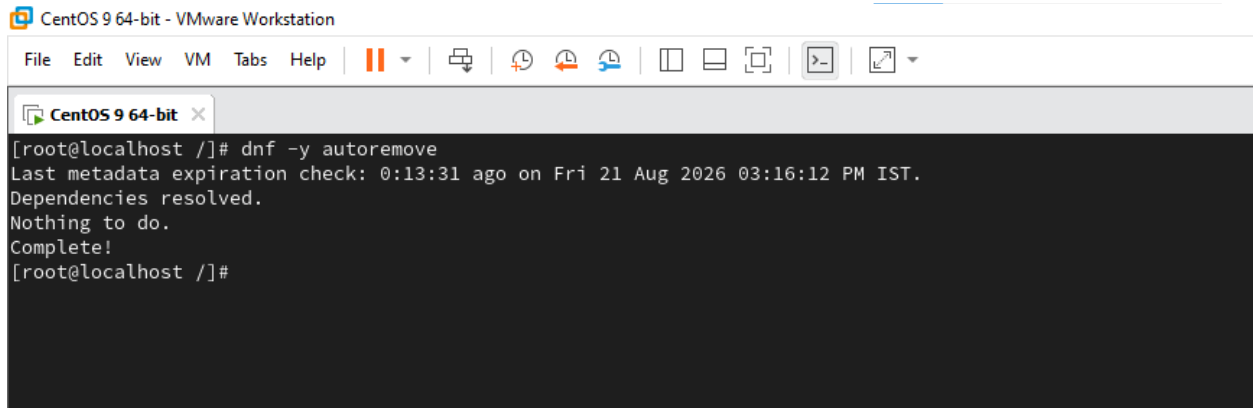
```
[root@localhost ~]# dnf list installed | grep -i python
libcap-ng-python3.x86_64                                0.8.2-7.el9 @AppStream
libpeas-loader-python3.x86_64                          1.30.0-4.el9 @AppStream
policycoreutils-python-utils.noarch                   3.6-2.1.el9 @AppStream
python-unversioned-command.noarch                     3.9.20-1.el9 @AppStream
python3.x86_64                                          3.9.20-1.el9 @anaconda
python3-audit.x86_64                                   3.1.5-1.el9 @AppStream
python3-brlapi.x86_64                                  0.8.2-4.el9 @AppStream
python3-cairo.x86_64                                   1.20.1-1.el9 @AppStream
python3-chardet.noarch                                 4.0.0-5.el9 @anaconda
python3-cups.x86_64                                    2.0.1-10.el9 @AppStream
python3-dasbus.noarch                                  1.4-5.el9 @AppStream
python3-dateutil.noarch                                1:2.8.1-7.el9 @anaconda
python3-dbus.x86_64                                    1.2.18-2.el9 @anaconda
python3-distro.noarch                                  1.5.0-7.el9 @AppStream
python3-dnf.noarch                                     4.14.0-18.el9 @anaconda
python3-dnf-plugins-core.noarch                       4.3.0-16.el9 @anaconda
python3-file-magic.noarch                             5.39-16.el9 @AppStream
python3-firewall.noarch                               1.3.4-20.el9 @baseos
python3-gobject.x86_64                                 3.40.1-6.el9 @AppStream
python3-gobject-base.x86_64                           3.40.1-6.el9 @anaconda
python3-gobject-base-noarch.noarch                    3.40.1-6.el9 @anaconda
python3-gpg.x86_64                                     1.15.1-6.el9 @anaconda
python3-hawkey.x86_64                                  0.69.0-12.el9 @anaconda
python3-ldna.noarch                                    2.10-7.el9.1 @anaconda
python3-libcomps.x86_64                                0.1.18-1.el9 @anaconda
python3-libdnf.x86_64                                  0.69.0-12.el9 @anaconda
python3-libs.x86_64                                    3.9.20-1.el9 @anaconda
python3-libseltlinux.x86_64                           3.6-2.el9 @AppStream
python3-libseltlinux.x86_64                           3.6-2.el9 @AppStream
python3-libstoragegmt.x86_64                          1.10.1-1.el9 @AppStream
python3-libxml2.x86_64                                 2.9.13-6.el9 @anaconda
python3-linux-procfs.noarch                           0.7.3-1.el9 @anaconda
python3-louis.noarch                                  3.16.1-5.el9 @AppStream
python3-lxml.x86_64                                    4.6.5-3.el9 @AppStream
python3-nftables.x86_64                               1:1.0.9-3.el9 @anaconda
python3-perf.x86_64                                   5.14.0-514.el9 @anaconda
python3-pexpect.noarch                                4.8.0-7.el9 @anaconda
python3-pip-wheel.noarch                              21.3.1-1.el9 @anaconda
python3-policycoreutils.noarch                        3.6-2.1.el9 @AppStream
python3-psutil.x86_64                                 5.8.0-12.el9 @AppStream
python3-ptyprocess.noarch                             0.6.0-12.el9 @anaconda
python3-pyatspi.noarch                                2.38.1-3.el9 @AppStream
python3-pycurl.x86_64                                 7.43.0-6-8.el9 @AppStream
python3-ovsocks.noarch                                1.7.1-12.el9 @anaconda
```

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Q.5 Use ChatGPT or Copilot to generate the command needed to clean up unused packages and dependencies on your Linux system, then run the command and share the result. Hint: Ask the AI tool: 'What is the command to remove unused packages in Ubuntu/Debian?' and try it on your system.

ANS :



The screenshot shows a terminal window titled 'CentOS 9 64-bit - VMware Workstation'. The terminal output is as follows:

```
[root@localhost ~]# dnf -y autoremove
Last metadata expiration check: 0:13:31 ago on Fri 21 Aug 2026 03:16:12 PM IST.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost ~]#
```